The evolution of blocking: a reinforcement learning account

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In linguistic blocking phenomena, a specific term blocks the use of a general one, leading to a more efficient one-to-one mapping between forms and meanings (Horn, 1984). The evolution of blocking is modeled here with a simple reinforcement learning set up, which moreover predicts the strength of the blocking from the relative frequency of the alternative meanings. The account is illustrated with an analysis of the rising use of reflexive pronouns in the history of English.

1. Introduction

English reflexive pronouns such as *herself* require a local antecedent, while simple pronouns such as *her* systematically reject a local antecedent.[1] So *herself* in 1a refers to Mary while *her* in 1b can refer to any female individual except Mary:

(1) a. Mary$_i$ admires herself$_{i/*j}$. *conjoint*
b. Mary$_i$ admires her$_{j/*i}$. *disjoint*

The *self*-forms are said to have the CONJOINT reference, while the simple pronouns have DISJOINT reference. The conjoint reading of *herself* is said to BLOCK that reading of *her*. Indeed, simple pronouns allowed conjoint readings in Old English, until the *-self* forms took over that function. Blocking results in a more efficient one-to-one mapping between forms (*herself, her*) and interpretations (conjoint, disjoint). So it is important to understand how blocking arises, in English anaphoric binding and the other grammatical phenomena where it is observed (Horn, 1984).

Blocking of a general term by a more specific one is common but hardly ubiquitous. What sort of conditions give rise to it? We addressed this question by modeling the evolution of blocking in English pronominal anaphora, within a variant of reinforcement learning that we call SEQUENTIAL REINFORCEMENT LEARNING (Wechsler, Shearer, & Erk, 2025; Boguraev, Erk, Mahowald, Shearer, & Wechsler, 2025; Shearer, 2025).

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Using numerical simulations within our evolutionary model, we arrived at the following main result: blocking becomes obligatory if the blocking form (*herself*) has a meaning (conjoint) that is rarer than the alternative (disjoint). A corpus study confirms that that it is rarer, thus predicting the rise in blocking.

1.1. *The history of blocking in English anaphoric binding*

The pattern in (1) has held in English since the late 1700s (Stage III in Table 1). Let us consider stages of English leading up to this state. In Stage I the unmarked pronouns had both disjoint and conjoint interpretation (Keenan, 2002 6a, 16a):

(2) syðÞan he hine to guðe gegyred hæfde
once he.NOM$_i$ him.ACC$_i$ for battle girded had
'once he had girded himself (lit. 'him') for battle' (c750, Beowulf 1473)

In a usage still found today, the *self*-forms were sometimes used appositionally as markers of surprisal: *hwæt Crist self tæhte* 'what Christ himself taught' (c1000; ÆO & N Pref). Then *self*-forms in object positions were reanalyzed as reflexives:

(3) Þe tre[sur] Þat godd ȝef him seolf fore
the treasure that God$_i$ gave [him self]$_i$ for
'the treasure that God gave himself for' (c1200; Sawles Warde)

However, conjoint readings of ordinary pronouns, as in 2, persisted during this stage, so the two forms were alternating (Stage II in Table 1).

Table 1, right, gives frequency counts of the two forms in conjoint readings, from a corpus study by Keenan, 2002. In Old English corpora, only 18% of conjoint object pronouns are *self*-forms. The share of *self*-forms grew larger in the 1500s. By the 1700s there were very few conjoint readings of direct object simple pronouns, as they were apparently blocked by the reflexives.

Table 1. Left: Stages in the rise of English *self*-forms. Right: Frequency of simple pronouns vs. *-self* forms for conjoint meaning. Data from Keenan, 2002.

	pron.	*-self*
Stage I	*disjoint* ∨ *conjoint*	
Stage II	*disjoint* ∨ *conjoint*	*conjoint*
Stage III	*disjoint*	*conjoint*

	Pron	*-self*	%self
c750-1154	419	89	18%
1154-1303	735	167	19%
1303-1400	753	191	20%
1400-1495	915	203	18%
1495-1605	291	1232	81%
1605-1700	138	930	87%
1722-1777	3	335	99%

1.2. *Sequential Reinforcement Learning Theory*

We addressed the evolution of blocking through numerical simulations of language histories using SEQUENTIAL REINFORCEMENT LEARNING (SRL; Wech-

sler et al., 2025; Boguraev et al., 2025; Shearer, 2025). A simple learning/production algorithm models the speaker's choices, thereby generating language histories. In SRL learner/speakers seek to match the input language they have previously witnessed, including self-witnessing. Applied to the current problem this means concretely that a speaker at a given stage in the history of English, expressing conjoint reference as in 1a ('Mary admires herself') is more likely to choose the form *her* for the object, the more likely it is to express the desired message, as determined by frequencies in the input (see formula (4)). If they reject the form *her*, then *herself* is the only alternative (see formula (5)). Although speakers seek to match the input, the frequency of *herself* nonetheless rises in this model, as we shall see, matching the history of English (Table 1, right). We discuss the reason it works and compare SRL to other learning models in Sec. 4 below.

2. Methods

We simulated language histories with a simple production set-up. First a speaker chooses between a conjoint and disjoint message, from a fixed probability distribution. If the disjoint message is chosen, they produce an utterance with the simple pronoun, as there is no alternative. If the conjoint message is chosen, the speaker must make a choice as to whether to use the simple pronoun or the *self*-form. Our key model assumption is that the speaker chooses the simple pronoun in proportion to how often the simple form has expressed the conjoint message before, as opposed to expressing the disjoint message.

In our model, every time the speaker makes this choice, counts are updated. All disjoint clauses use the simple pronoun, and the use of simple pronoun for conjoint is discouraged by the many simple pronouns used for the unintended disjoint meaning. As a result, the count of simple pronouns in conjoint use grows more slowly than those in disjoint.

Estimating message probabilities To estimate the relative frequency of disjoint versus conjoint objects in English, we counted pronouns in accusative case immediately following a verb, in the Corpus of Contemporary American English (www.english-corpora.org/coca). We counted simple pronouns *me, us, him, her, them* and *self*-forms *myself, ourselves, himself, herself, themselves* separately. There were 4,932,323 simple pronouns and 418,276 self-forms.

The COCA data are from Modern English, when the blocking effect is nearly total. So we assume the proportion of simple versus self-forms directly reflects the proportion of disjoint versus conjoint messages. Based on this method, we estimated that 92.2% of object pronominals are disjoint, while 7.8% are conjoint. For the purpose of this study we assume that earlier stages of the language have a similar distribution of meanings.

Reinforcement learning model In reinforcement learning, behaviors with positive outcomes are reinforced (Roth & Erev, 1995; Harley, 1981). In the present study, we use use a variant of the Harley-Roth-Erev formula from Wechsler et al., 2025 and Boguraev et al., 2025. If there are n different messages $m_1 \ldots m_n$ expressed by form f, with previous counts $c_1 \ldots c_n$, then the probability of uttering f to express meaning m_i is: $p(f|m_i) = \frac{c_i}{\sum_j c_j}$. As we will see, a "rich get richer" effect is observed in the present model of English reflexive pronouns due to the uneven distribution of messages.

Present study Our toy 'Object Pronoun Language' has the three sentences shown in Table 2. A speaker prepares to make an utterance by first randomly selecting a message, either message D ('Disjoint') or K ('Konjoint'), proportional to $p(D)$. If they select D they utter sentence (a). If they select K they utter either (b) or (c), a choice determined by the production algorithm below.

Table 2. The Object Pronoun Language. Key to symbols for messages expressed, forms, and counts of previous utterances.

sentence	message	form	counts
a. Mary$_i$ admires her$_j$.	D	$f^{\emptyset}$	$c_D^{\emptyset}$
b. Mary$_i$ admires her$_i$.	K	$f^{\emptyset}$	$c_K^{\emptyset}$
c. Mary$_i$ admires herself$_i$.	K	f^{SELF}	c_K^{SELF}

The production algorithm generates a different set of language histories depending on the message probability distribution $p(D) + p(K) = 1$. Below we present the results of simulations using values of $p(D)$ and $p(K)$ from English, as well as other hypothetical values. The model keeps track of three counts from previous utterances: form $f^{\emptyset}$ in disjoint meaning, and $f^{\emptyset}$ and f^{SELF} in conjoint meaning; these are notated $c_D^{\emptyset}$, $c_K^{\emptyset}$ and c_K^{SELF}, respectively (D for disjoint, K for 'konjoint'). Those counts are used in the following formulae, when speaker chooses between $f^{\emptyset}$ and f^{SELF}:

(4) $$p(f^{\emptyset} \mid K) = \frac{c_K^{\emptyset}}{c_D^{\emptyset} + c_K^{\emptyset}}$$

(5) $$p(f^{\text{SELF}} \mid K) = 1 - p(f^{\emptyset} \mid K) = \frac{c_D^{\emptyset}}{c_D^{\emptyset} + c_K^{\emptyset}}$$

For simplicity, we assume that speakers observe all speech acts in the community, so that we only need to keep one set of counts for the whole speech community.

The learning/production algorithm has three steps:

Step 1 The speaker selects a message, D or K, proportional to $p(D)$ (or, equivalently, $p(K)$ since this is exactly $1 - p(D)$).

Step 2 (i) If they selected message D, then they utter $f^{\emptyset}$ (Table 2, (a)), as this is their only option. They utter sentence (a), and proceed to Step 3.
(ii) If they selected message K, they choose between forms $f^{\emptyset}$ and f^{SELF}, using counts of previous utterances in the language history (including their own earlier utterances). The learning formulae produce a probability distribution over $f^{\emptyset}$ and f^{SELF}, from which speakers select a form, and utter (b) or (c) from Table 2.

Step 3. Update the counts: if (a) (= $f^{\emptyset}_{D}$) is uttered, add 1 to $c^{\emptyset}_{D}$; if (b) (= $f^{\emptyset}_{K}$) is uttered add 1 to $c^{\emptyset}_{K}$; if (c) (= f^{SELF}_{K}) is uttered add 1 to c^{SELF}_{K}. Then return to Step 1 and repeat.

Numerical simulations We carried out numerical simulations, computing 10,000 language histories out to 100,000 utterances, for each test. The output of a simulation is a sequence of 'utterances' of sentences (a), (b), and (c) from Table 2. We varied the message probabilities as a parameter. In the first test we used the message probabilities from our corpus study, namely 92.2% disjoint and 7.8% conjoint. In the second test we simulated a hypothetical language in which the relative probabilities are reversed: 25% disjoint and 75% conjoint.

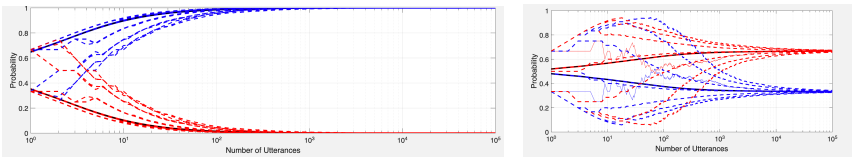

Figure 1. Log scale plots of simulated language histories for the Object Pronoun Language. The plots reflect counts of conjoint messages only: red represents the proportion expressed with simple pronouns (*her*), blue represents the proportion expressed with reflexive pronouns (*herself*). Thick lines: the mean; dashed lines: percentiles (0.1%, 1%, 10%, 50%, 90%, 99%, and 99.9%). **Left**: Realistic English message probabilities: 92.2% disjoint, 7.8% conjoint. **Right**: A hypothetical language where the marked meaning is more frequent than unmarked; message probabilities 25% disjoint, 75% conjoint.

3. Results

If the unmarked meaning is the more frequent of the two, the marked form will rise in frequency until it completely blocks the unmarked form. The plot in Fig. 1 (left) shows the results for unmarked at 92.8% and marked at 7.8%: the probability of the marked self-form converges to one (blue plots), and that of the simple pronoun converges to zero (red plots).

Fig. 1, left, shows the results of 10,000 language history scenarios, plotted on a log scale. For this simulation we used the estimated message probabilities for

English object pronouns, namely 92.2% disjoint and 7.8% conjoint. The plots represent the changing probability that a speaker expressing a conjoint message will choose each type of pronoun: blue plots for the reflexive (*herself*), red plots for the simple pronoun (*her*). The thick lines represent the means across all 10,000 simulations; dashed lines represent percentile values: 0.1%, 1%, 10%, 50%, 90%, 99%, and 99.9%. The main result is that the probability that the speaker will utter a reflexive pronoun converges to 1, while forms such as *Mary*$_i$ *admires her*$_i$ (meaning 'Mary admires herself'; cp. Old English (2)), drop out of the language. Wechsler et al., 2025 showed through analytic means that (i) the probability of the marked form always converges to 1, as long as the unmarked meaning is more frequent than the marked (here, as long as $p(D) > p(K)$); and through numerical simulations that (ii) convergence is faster, the greater the difference between the two. In fact the marked meaning is far rarer—only 7.8% of the total—and convergence is rapid (Fig. 1, left).

Suppose instead that the marked meaning were more common than the unmarked: what would the model predict? A sample answer is shown in Fig. 1 right, where we assumed that only 25% of messages have the unmarked meaning, while 75% are have the marked. The result is that it converges to a constant value less than one, in this case 1/3 (.33) *self*-forms and 2/3 (.66) simple pronouns. Wechsler et al., 2025 showed through analytic means that with parametric probability p, as the number of utterances grows large, the probability of choosing the *unmarked* form converges to $(1-2p)/(1-p)$. In our example, $p = \frac{1}{4}$ messages are disjoint, so the predicted frequency of simple pronouns is $(1-2(\frac{1}{4}))/(1-\frac{1}{4}) = \frac{2}{3}$, which is what we see in Fig. 1b. In ongoing work, this kind of SRL model is applied to the analysis of differential object marking (optional accusative case) in Hindi (Deshpande & Wechsler, 2026).

4. Discussion

Why does the SRL model work? What are the intuitions behind the math? Our learning formulas share two important features with those of classic reinforcement learning (Harley, 1981; Roth & Erev, 1995): (i) the probability that an agent takes a certain action (utters a certain expression) depends on how many times they have witnessed similar actions (utterances) in the past; and (ii) that count serves as both the numerator and as one addend in the denominator (see 4). If that count grows more quickly than the remaining addends then the fraction rises and converges to 1 in the limit. The same principle is at work in the earlier RL models. However, comparing to other language learning models, our formulae specifically resemble those of the 'obverter model' (Oliphant & Batali, 1997; Spike, Stadler, Kirby, & Smith, 2017), in the following respect: the addends in the denominator (e.g. $c_D^{\emptyset}$ and $c_K^{\emptyset}$ in formula (4)) are not counts of alternative *forms* but rather alternative *meanings*. A speaker decides about a form based on the distribution of meanings expressed by previous productions of that form. Oliphant and Batali (1997)

conceive of this as a speaker 'obverting' the planned utterance in order to see it from the perspective of the hearer who must determine the meaning. We see it as a speaker seeking to match the input by consulting their own memories of producing and hearing previous utterances. So our model doesn't require any actual reasoning by the speaker about the hearer, it just requires speakers/hearers to keep track of observed frequencies.

As a type of language use, blocking appears to involve the speaker and hearer taking each other's perspective. Intuitively, a cooperative speaker with a conjoint message should avoid ambiguity out of consideration for the hearer. Meanwhile, a hearer interpreting a simple pronoun reasons that a cooperative speaker would probably have used the unambiguous form if they intended the conjoint meaning. In addition to the obverter model, this dynamic has inspired approaches such as neo-Gricean ones (Horn, 1984), bi-directional Optimality Theory (see Beaver & Lee, 2004 for a critique), signaling game theory (Deo, 2015; Franke & Jäger, 2012), Rational Speech Act theory (see Dionne & Coppock, 2022 for an RSA treatment of blocking), and iterated learning with an 'avoid ambiguity' bias (Kirby, Tamariz, Cornish, & Smith, 2015).

While these approaches capture something of the dynamic of language use, they are problematic as motivations for language change. The problem is simply that grammatical systems are clearly quite tolerant of ambiguity (Piantadosi, Tily, & Gibson, 2012). Speakers of Old English were not saddled with a sub-optimal language that was slowly improved over the centuries. Instead, people disambiguate expressions based on the context of utterance. The optimization schemes above can be seen as providing rational approaches to disambiguation in the absence of context.

5. Conclusion

In this paper we have provided an explanation for the emergence of blocking based on a simple learning model. Other SRL analyses to date include studies of the rise of the English progressive (Wechsler et al., 2025) and typological patterns of NP split ergativity (Boguraev et al., 2025). The central assumptions are (i) that a speaker is more likely to utter a form if they believe it is more likely to convey their intended message, and (ii) speakers estimate those likelihoods from the input. The theory also makes rather specific model predictions relating the nature and speed of convergence to the relative frequency of different messages, which we were able to estimate empirically. We look forward to more work in this approach.

References

Beaver, D., & Lee, H. (2004). Input-output mismatches in optimality theory. In R. Blutner & H. Zeevat (Eds.), *Optimality theory and pragmatics* (pp. 112–153). London: Palgrave Macmillan.

Boguraev, S., Erk, K. E., Mahowald, K., Shearer, J. W., & Wechsler, S. M. (2025). Reinforcement learning produces efficient case-marking systems. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 47).

Deo, A. (2015). The semantic and pragmatic underpinnings of grammaticalization paths: The progressive to imperfective shift. *Semantics and Pragmatics*, *8*(14), 1–52.

Deshpande, S., & Wechsler, S. (2026). *Differential object marking in Hindi: an Evolutionary LFG account.* (unpublished abstract, The University of Texas)

Dionne, D., & Coppock, E. (2022). Complexity vs. salience of alternatives in implicature: A cross-linguistic investigation. *Glossa Psycholinguistics*, *DOI: 10.5070/G601190*.

Franke, M., & Jäger, G. (2012). Bidirectional optimization from reasoning and learning in games. *Journal of Logic, Language and Information*, *21*, 117–139.

Harley, C. B. (1981). Learning the evolutionarily stable strategy. *Journal of Theoretical Biology*, *89*(4), 611–633.

Horn, L. (1984). Toward a new taxonomy for pragmatic inference: Q-based and R-based implicature. In D. Schiffrin (Ed.), *Meaning, Form, and Use in Context: Linguistic Applications* (pp. 11–42). Georgetown: Georgetown University Press.

Keenan, E. L. (2002). Explaining the creation of reflexive pronouns in English. In D. Minkova & R. Stockwell (Eds.), *Studies in the History of the English Language: A Millennial Perspective* (pp. 325–354). Berlin, New York: De Gruyter Mouton.

Kirby, S., Tamariz, M., Cornish, H., & Smith, K. (2015). Compression and communication in the cultural evolution of linguistic structure. *Cognition*, *141*, 87–102.

Oliphant, M., & Batali, J. (1997). Learning and the emergence of coordinated communication. *Center for Research on Language Newsletter*, *11*(1).

Piantadosi, S. T., Tily, H., & Gibson, E. (2012). The communicative function of ambiguity in language. *Cognition*, *122*(3), 280–291.

Roth, A. E., & Erev, I. (1995). Learning in extensive-form games: Experimental data and simple dynamic models in the intermediate term. *Games and Economic Behavior*, *8*(1), 164–212.

Shearer, J. W. (2025). Coordination games with sequential stochastic learning and language emergence. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 47).

Spike, M., Stadler, K., Kirby, S., & Smith, K. (2017). Minimal requirements for the emergence of learned signaling. *Cognitive Science*, *41*(3), 623–658.

Wechsler, S., Shearer, J. W., & Erk, K. (2025). The emergence of grammar through reinforcement learning. *arXiv preprint arXiv:2503.01635*.